Chapter 2 – Literature Review

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2.1 Introduction

A literature review is essential to understand how similar problems have been addressed and what design approaches previous systems have adopted. Reviewing existing studies and systems related to task allocation, skill matching, and automation provides insight into current technologies, their strengths, and their limitations. This understanding helps identify research gaps and justifies the need for the Intelligent Task Allocation System (ITAS). By analyzing prior work, the project can build upon proven concepts while avoiding known shortcomings.

2.2 Review of Existing Systems

System 1: Jira Smart Assignment and Automation Tools This system focuses on automating task assignment by utilizing predefined rules such as task type, priority levels, and workload balancing.

- **Strengths:** Its primary strength lies in its ability to significantly reduce manual administrative effort and improve operational efficiency. It excels at managing general workflow distribution and ensuring that no single team member is overwhelmed with volume.
- **Weaknesses:** However, the system is limited by its inability to analyze employee skills in depth. It does not extract or process data from CVs or portfolios, meaning assignments are based on static rules rather than actual technical competency or suitability.
- **Difference from Proposed System:** Unlike Jira, which relies on manual configuration and simple capacity rules, the proposed ITAS solution actively analyzes the semantic content of employee CVs and portfolios. This allows ITAS to assign tasks based on verified technical skills and suitability scores rather than just availability or predefined rule sets.

System 2: GitHub Projects and GitHub Actions GitHub combines project management with automation actions to manage issues and tasks directly within the software development environment. Tasks are routed using labels, templates, or keyword-based rules.

- **Strengths:** The major strength of this approach is its high flexibility and seamless integration into the developer's existing workflow. It allows for immediate, code-centric task management without switching tools.
- **Weaknesses:** Its weakness is that it focuses almost entirely on workflow automation (moving tickets) rather than intelligent resource allocation. It lacks the capability to systematically match tasks to individuals based on their specific skill suitability, experience level, or historical performance.
- **Difference from Proposed System:** While GitHub automates the mechanics of moving tasks through a workflow, ITAS introduces an intelligence layer that evaluates *who* is best fit to perform the work. ITAS specifically targets the "knowledge gap" by extracting skills from academic and portfolio data to surface talent that manual GitHub assignments might overlook.

System 3: NLP-Based Applicant Tracking Systems (ATS) These systems are widely used in recruitment to match job descriptions with candidate resumes by extracting skills and keywords to compute similarity scores.

- **Strengths:** They effectively demonstrate the power of Natural Language Processing (NLP) for parsing text and identifying relevant qualifications, providing a strong foundation for understanding how to match documents to requirements.
- **Weaknesses:** The critical weakness is their context; they are designed exclusively for hiring decisions (a one-time event). They are not adapted for the dynamic environment of active projects, meaning they cannot handle ongoing task allocation, changing project requirements, or daily workload management.
- **Difference from Proposed System:** The proposed system adapts the core NLP matching technology found in ATS but repurposes it for internal project management. Unlike an ATS, which stops functioning once a candidate is hired, ITAS continues to monitor the employee's workload via a dashboard and allocates specific project tasks based on real-time availability.

System 4: Smart Hiring and CV Matching Systems Smart hiring platforms apply advanced techniques, such as Named Entity Recognition (NER) and embedding-based similarity, to match candidates with job requirements with a focus on explainability.

- **Strengths:** These systems offer advanced matching capabilities that are transparent and highly accurate in identifying the best candidate for a role based on complex criteria.
- **Weaknesses:** Despite their sophistication, they are not designed to manage the lifecycle of a software task. They lack features for monitoring real-time workloads, tracking task progress, or continuously allocating small-scale assignments within a project team.
- **Difference from Proposed System:** While Smart Hiring systems focus on the initial recruitment of a candidate, ITAS focuses on the operational utilization of that candidate within a team. ITAS integrates matching logic directly with project execution, ensuring that the skills identified during hiring are accurately mapped to daily development tasks to prevent the underutilization of talent.

System 5: Competency Profiling and Skill Management Systems These systems are dedicated to building detailed inventories of employee skills, certifications, and competencies. They are often used by HR departments to map the organization's capabilities and identify skill gaps for training purposes.

- **Strengths:** They provide excellent visibility into the theoretical capabilities of a workforce. They allow organizations to maintain a structured database of who knows what, which is useful for long-term strategic planning and employee development.
- **Weaknesses:** A major limitation is that these profiles are often static and require manual updating by the employees or managers. They are rarely integrated with daily workflow tools, meaning the "profile" often becomes outdated and does not reflect the practical, hands-on experience gained from recent tasks.
- **Difference from Proposed System:** Competency profiling systems act as passive databases, whereas ITAS is an active allocation engine. The proposed system does not just store skill data; it dynamically uses that data to drive daily operations. Furthermore, ITAS automates the creation of these profiles from raw data (CVs/Portfolios) rather than relying on manual data entry, ensuring the skill information is derived from verified sources rather than self-reported claims.

2.3 Comparison of Existing Solutions

Existing systems vary in their goals and technical approaches. Task management tools focus on workflow efficiency, while recruitment systems emphasize candidate-job matching. None of the reviewed solutions fully integrate skill extraction, fair task allocation, workload balancing, and performance feedback into a single system.

System	Main Purpose	Skill-Based Matching	Workload Balancing	Key Limitation
Jira Smart Assignment	Task automation	No	Yes	No deep skill analysis
GitHub Projects	Workflow automation	Limited	Indirect	No suitability scoring
NLP-based ATS	Candidate ranking	Yes	No	Not task-focused
Smart Hiring Systems	CV-job matching	Yes	No	No project task assignment

The analysis highlights a clear gap in existing solutions: the lack of an integrated system that allocates tasks based on skill suitability while considering workload, fairness, and continuous performance improvement. ITAS aims to address these gaps by combining intelligent skill matching, task management, and feedback mechanisms into a unified framework.

Updated References (2021–2025)

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